Categorical Encoding and Threshold Choice for BRFSS Classification

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***Abstract*—We classify self-reported myocardial infarction or coronary heart disease in an imbalanced BRFSS-derived survey dataset. Ten NumPy linear and logistic configurations compare imputation, missing indicators, categorical encoding and regularization on separate development and tuning partitions. At a common threshold, categorical encoding increases tuning F1 by 0.0147; a development-only fold check finds positive differences in all three folds. The frozen regularized logistic model obtains holdout F1 0.4225, accuracy 0.8685 and recall 0.5446. Threshold choice substantially changes this tradeoff. Optimization remains budget-limited, while survey structure and self-report prevent interpreting these results as prospective clinical validation.**

I. Data and experimental design

We study how imputation, categorical representation, regularization and threshold choice affect minority-class classification in an imbalanced survey. The target `_MICHD` concerns reported history of myocardial infarction or coronary heart disease, rather than verified future disease onset. The supplied data contain 328,135 labeled rows and 109,379 test rows with 321 features. There are 28,975 positives (8.8302%). Predicting every row negative therefore obtains 0.9117 accuracy, but no true positives, zero recall and zero F1.

We report unweighted sample-level classification. BRFSS uses telephone sampling, design weights and raking [2]; neither our loss nor our evaluation uses those weights as observation weights. Sample prevalence and performance are consequently not design-adjusted population estimates. Raw missingness is 44.7880% in training and 44.7852% in test. Schemas and label/test IDs align. No exact duplicate rows were found across all raw columns, including survey keys; this does not establish respondent-level independence.

Seed 433 defines a stratified split of 196,881 development, 65,627 tuning and 65,627 holdout rows. Imputation, scaling, categories, references and indicator inclusion use only the fitting partition, respecting the training-fold boundary [3]. The holdout was opened only after freezing the model and threshold. Subsequent precision and representation diagnostics use development data only.

F1 is our retained minority-class development objective. Written AIcrowd rules specify accuracy [4]; an observed leaderboard displaying F1 first does not establish a current staff clarification. Both metrics are reported; F1 is not claimed as the confirmed competition metric.

II. Representation and learning

Preprocessing. We exclude `Id`, sequence number `SEQNO`, its duplicate `_PSU`, and raw date `IDATE`. We also exclude `HAREHAB1`, a rehabilitation question with heart-attack-dependent availability [1]. All 773 nonmissing training entries belong to positives. This specific proxy check is not an exhaustive audit of every survey module.

Numeric columns receive fitted mean or median fill and fitted mean/std scaling; fully observed constants and all-missing columns are removed. Optional indicators encode missingness observed during fitting. The categorical variant reference-encodes integer columns with at most twelve observed fitting levels. The most frequent level is the reference, with ties favoring the lowest value; unseen levels map to the reference. This increases flexibility and dimensionality without imposing ordinal spacing. Higher-cardinality nominal codes can still remain numeric. We do not claim complete semantic cleaning: `PHYSHLTH`=88, for example, means zero unhealthy days [1], so universal sentinel replacement is unjustified.

Models. The six required methods retain the specified signatures, one-dimensional weights and final-state data losses. Required ridge solves $(X^\top X+2N\lambda I)w = X^\top y$, penalizing every supplied coordinate. The optional linear baseline uses a Gram eigen pseudoinverse for redundancy; positive ridge uses a float64 solve. For optional logistic prediction, with X of shape $N \times D$, $y_i \in \{0, 1\}$ and intercept w_0, we minimize

$$J(w) = \frac{1}{N}\sum_{i=1}^{N}[\log(1 + e^{x_i^\top w}) - y_i x_i^\top w] + \lambda \sum_{j=1}^{D-1} w_j^2. \quad (1)$$

Its intercept convention is separate from the mandatory methods. NumPy L-BFGS uses ten curvature pairs and Armijo backtracking. The historical fit stores float32 designs and rounds optimization matrix–vector products to float32, with float64 weights and reductions. That arithmetic is preserved for reproduction; a separate blocked float64 path passes finite-difference checks. The frozen budget is 120 updates and the requested gradient infinity norm is 10^{-5}. The all-label final norm is 0.0002108, so convergence is not established.

Selection. Ten configurations compare preprocessing and model families. The categorical design uses $\lambda = 0$, 10^{-4}, 10^{-3}, 10^{-2} or 10^{-1}. Thresholds span 0.05–0.65 in steps of 0.01 on tuning predictions only; equal metric values prefer higher thresholds. Within 0.002 of maximum tuning F1, the

Table I
TUNING COMPARISONS WITH INDIVIDUALLY TUNED THRESHOLDS. LINEAR MODELS USE THE INDICATOR DESIGN. THE MAJORITY CLASSIFIER IS FIXED.

Model	Threshold	Accuracy	F1	Recall
Majority	—	0.9117	0.0000	0.0000
Linear regression	0.22	0.8653	0.4057	0.5208
Ridge (10^{-3})	0.22	0.8666	0.4049	0.5141
Logistic, mean fill	0.21	0.8724	0.4133	0.5089
Categorical, no L2	0.21	0.8763	0.4327	0.5343
Selected L2 logistic	0.20	0.8722	0.4321	0.5506

Table II
TUNING ABLATIONS AT COMMON THRESHOLD 0.20. L2 IS 10^{-3} UNTIL THE FINAL ROW. DELTAS ARE ABSOLUTE F1 UNITS. EQUAL BUDGETS DO NOT IMPLY EQUAL CONVERGENCE.

Intervention	F1	ΔF1
Mean fill + scaling	0.4129	—
Median fill	0.4131	+0.0002
+ Missing indicators	0.4139	+0.0008
+ Categorical encoding	0.4285	+0.0147
L2 $10^{-3} \to 10^{-4}$	0.4321	+0.0036

recorded rule prefers fewer columns, then stronger regularization, then higher F1. It selects $\lambda = 0.0001$, threshold 0.20 and 1,172 columns. This tolerance is a deliberate preference, not evidence of superior generalization.

III. RESULTS AND INTERPRETATION

Table I uses each model's tuning-selected threshold. The unregularized categorical candidate reaches 0.4327 tuning F1, slightly above the selected model's 0.4321, within the stated tolerance. Table II instead fixes the threshold at 0.20. Categorical coding gives the largest representation gain; imputation and indicator changes have much smaller effects. These are tuning comparisons, not causal or statistically significant improvements. A subsequent numeric-versus-categorical check on three shared development-only folds, with $\lambda = 10^{-3}$ and the same budget and threshold, gives positive F1 differences from 0.0084 to 0.0168. This supports repeatability on reused data, not independent evaluation of the full search. Increasing L2 to 10^{-2} lowers tuned F1 to 0.4099.

On the frozen holdout, F1 is 0.4225, accuracy 0.8685, balanced accuracy 0.7223, precision 0.3451 and recall 0.5446. Counts are TP=3,156, FP=5,989, FN=2,639 and TN=53,843; ROC-AUC is 0.8585 and average precision 0.3861. At threshold 0.50, the same scores give F1 0.2444, accuracy 0.9142 and recall 0.1572. The tuning-selected threshold adds 0.1781 holdout F1 while losing 0.0456 accuracy. On the tuning set, selecting the threshold for accuracy gives 0.9153 accuracy and 0.1796 F1 at 0.58. The retained threshold remains 0.20.

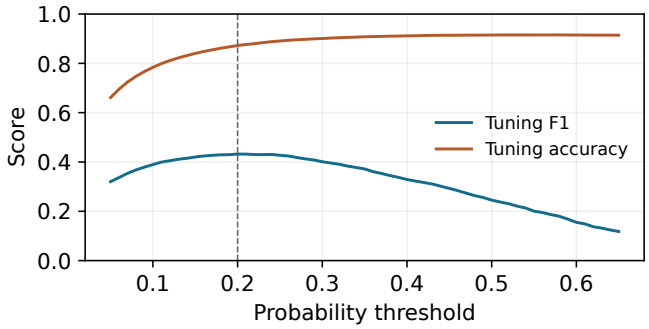


Figure 1. Original tuning scores for the selected model. The dashed line marks the tuning-selected threshold; holdout labels do not determine this choice.

Figure 1 illustrates the objective tradeoff. For calibrated conditional probabilities, the expected-F1-optimal threshold equals half the optimal F1 [5]. Calibration is not established here, so that special result neither determines nor validates our empirical threshold. Finite-sample selection can itself overfit [6]. The fixed configuration gives three-fold non-holdout F1 0.4310 $\pm$ 0.0004 (sample SD, not a standard error). Each fold refits processing and calibrates its threshold inside outer training, but family selection is not nested. A conditional IID-row bootstrap interval for holdout F1 is [0.4125, 0.4321]; it omits survey design and selection variability.

IV. LIMITATIONS AND REPRODUCIBILITY

High missingness, incomplete coding semantics, survey structure and self-report limit generalization. False negatives remain 45.54% of positives, and increased recall introduces many false positives. Training F1 0.4392 versus tuning 0.4321 does not eliminate selection bias. A development-only float64 diagnostic reduced objective and gradient norm without a consistent F1 gain; the frozen model is retained. Clinical validity, population fairness and causal interpretation are unestablished.

After selection, all-label fitting discovers additional categories, giving 1,183 final columns. From the repository root, use `python run.py` with `--data-dir ../dataset`, `--output new.csv` and `--seed 433`. Test inputs and saved state suffice. The original CSV is reproduced exactly in Python 3.12.14 / NumPy 2.5.3 with four OpenBLAS threads, including a separate legacy refit; cross-environment bitwise identity is unverified. The frozen model and threshold are specified in `config/selected_model.json`; complete preprocessing state is stored with the fitted weights.

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